

Build A simple feed forward neural network .

Aim:

To build a simple feed forward neural network to recognize handwritten characters.

Objective :

1. To load and process the mnist dataset for neural network input.
2. To build feed forward neural network with hidden layers.
3. To train the model using stochastic gradient descent optimizer and sparse loss.
4. To predict the class of given handwritten image.

Pseudocode.

START

Load mnist dataset (training and testing data).

Flatten each image from 28×28 to 784 features.

Normalize pixel image to range $[0, 1]$.

Layer 1: Dense (128 neurons, Relu activation)

Layer 2: Dense (64 neurons, Relu activation)

Output Layer: Dense (10 neurons, Softmax activation)

epoch	loss
1	1.0547
2	0.3846
3	0.3246
4	0.2934
5	0.2689

Test Accuracy: 92.80%

3 - To train the model we use stochastic gradient descent optimizer and after 5 epochs

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Each image from 28x28 to 100x100

100x100 pixel image (28x28 to 100x100)

Compu model

optimize = stochastic gradient descent

Loss = sparse categorical cross entropy

metric = accuracy

Observation:

The loss decreases with each epoch. Shows that the model is learning.

Accuracy improves steadily during training.

Testing: overall accuracy of the model on the entire dataset.

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Result:

Successfully build a simple feed forward neural network to recognize handwritten characters.

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```
[3]: import torch
import torch.nn as nn
import torch.optim as optim
import torchvision
from torchvision import transforms
from torch.utils.data import DataLoader

[4]: transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize((0.5,), (0.5,))
])

[5]: train_dataset = torchvision.datasets.MNIST(
    root='./data', train=True, transform=transform, download=True
)
test_dataset = torchvision.datasets.MNIST(
    root='./data', train=False, transform=transform, download=True
)

[6]: train_loader = DataLoader(train_dataset, batch_size=64, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=64, shuffle=False)

[9]: import torch
import torch.nn as nn

class SimpleFFN(nn.Module):
    def __init__(self):
        super(SimpleFFN, self).__init__()
```

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Notebook Python 3 (ipykernel)

```
loss.backward()
optimizer.step()

running_loss += loss.item()

print(f"Epoch [{epoch+1}/{epochs}], Loss: {running_loss/len(train_loader):.4f}")

Epoch [1/5], Loss: 1.0547
Epoch [2/5], Loss: 0.3846
Epoch [3/5], Loss: 0.3246
Epoch [4/5], Loss: 0.2934
Epoch [5/5], Loss: 0.2689

[13]: print("Training finished")

Training Finished

[14]: correct, total = 0, 0
with torch.no_grad():
    for images, labels in test_loader:
        images, labels = images.to(device), labels.to(device)
        outputs = model(images)
        _, predicted = torch.max(outputs, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()

print(f"Test Accuracy: {100 * correct / total:.2f}%")

Test Accuracy: 92.80%
```

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